Honours Linear Algebra - Weeks 1-3 Review

Week 4 / No lecture

The same problem, three languages

$Ax = b$: equations / a combination of columns / a linear map $T(x) = Ax$.

Basis = independent + spanning. Existence of coordinates comes from spanning; uniqueness from independence. For $\dim V = n$, exactly n vectors: either property implies both. Count matters.

Rank–nullity. $A \in M_{m\times n}(K)$, rank r:

$$\dim\ker T = n - r, \qquad T \text{ injective} \iff r = n, \qquad T \text{ onto} \iff r = m.$$

For consistent $Ax = b$, all solutions are $x_p + \ker T$. A subspace only when $b = 0$.

Row reduction. $R = EA$, E invertible: same kernel, row space and column relations. For a column-space basis, use pivot columns of the **original** A.

One matrix ties it together

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \end{pmatrix} \sim R = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Pivots $1, 3$; $a_2 = 2a_1$. Thus

$$\ker A = \operatorname{Span}\left\{ \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} \right\}, \qquad \operatorname{Col}(A) = \operatorname{Span}\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\} = \mathbb{R}^2.$$

$T : \mathbb{R}^3 \to \mathbb{R}^2$ is onto, not injective. Check $3 = 1 + 2$.

$$Ax = \begin{pmatrix} 3 \\ 1 \end{pmatrix} \iff x = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} + t \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}, \quad t \in \mathbb{R}.$$

The two columns in the image are basis vectors; the null vector is a relation among all three columns. Different roles.

Direct sums; maps in coordinates

$V = U \oplus W$: every $v = u + w$ uniquely. Subtract two decompositions:

$$u - u' = w' - w \in U \cap W = \{0\}. \qquad \dim(U + W) = \dim U + \dim W - \dim(U \cap W).$$

For three summands, pairwise zero intersections do **not** suffice: the x-axis, y-axis and $y = x$ in $\mathbb{R}^2$.

Matrix columns are basis images:

$$[T]^{\mathcal{B}}_{\mathcal{C}} = ([T(b_1)]_{\mathcal{C}} \ \cdots \ [T(b_n)]_{\mathcal{C}}), \qquad [T(v)]_{\mathcal{C}} = [T]^{\mathcal{B}}_{\mathcal{C}}[v]_{\mathcal{B}}.$$

Composition: rightmost acts first. The intermediate coordinate basis must match.

Keep the hypotheses. Equal finite dimensions: injective $\iff$ onto. On $\mathbb{R}[t]$, differentiation is onto but kills constants. Also, $\dim_{\mathbb{R}} \mathbb{C} = 2$ while $\dim_{\mathbb{C}} \mathbb{C} = 1$: always fix the field.